Practical 6: Magmatism at constructive plate margins

On successful completion of this practical, you should be able to:

- Interpret the order of crystallization using a ternary phase diagram
- Use the lever rule to calculate proportions of solid and liquid on a ternary phase diagram
- Estimate the composition of plagioclase from its extinction angle
- Recognise and describe the textures of basaltic rocks in thin section
- Interpret the cooling history of basalts from mineralogical and textural observations

You will also practice the following skills:

- Convey key mineralogical and textural features of igneous thin sections using labelled sketches and diagrams

OPPORTUNITY FOR FORMATIVE FEEDBACK
Part 3 of this practical is an exam-style question. If you wish to receive formative feedback on your work in preparation for the practical test, you may hand in your thin section description at the end of the practical session, and I will return it to you next week with some brief written comments.

To make the most of the opportunity for feedback, allow yourself at least 60 minutes to work on Part 3. The rest of this practical is designed so that you can complete the questions at home if necessary.

Part 1 (15 minutes): Determining plagioclase composition from a thin section

You will need to refer to the Practical Reference Handout, p.27.

The composition of plagioclase can be determined by measuring the extinction angle between the two sets of twins. For the measurement to be accurate, the crystal must have a very specific orientation: the {010} twin planes must be exactly perpendicular to the plane of the thin section. Crystals with the correct orientation will have two properties:

1) When the twin plane is oriented N-S on the microscope cross-hairs, both twin planes will show the same light grey interference colour;
2) The clockwise angle of rotation needed to send one set of twins into extinction and anticlockwise angle needed to send the other set of twins into extinction must be within <5° of each other.

Finding an appropriate crystal in a thin section involves a lot of looking around, and a lot of trial and error – so I have provided the questions below guide you through the process of determining plagioclase composition using some synthetic examples and data. Once you have got the hang of the method, you will be able to deploy this technique on real thin sections.

Which of the following plagioclase crystals are in an appropriate orientation to measure the extinction angles of the twin planes? (Indicate YES or NO)

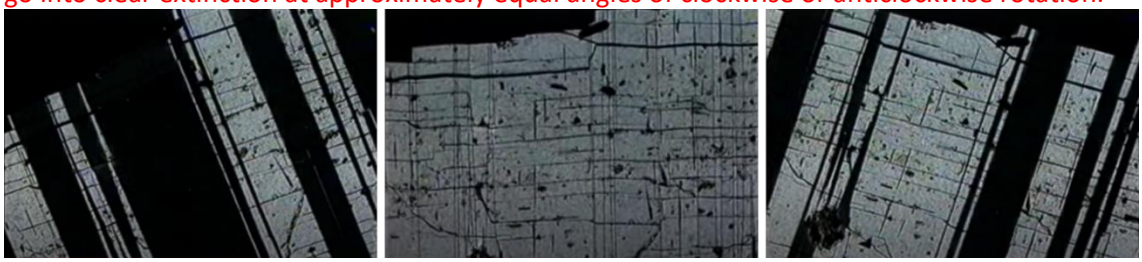
Crystal 1 YES this is a good candidate crystal – the twins are an equal shade of grey when they are oriented N-S on the cross-hairs.



Crystal 2 NO, not appropriate – the twins are not an equal shade of grey when they are oriented N-S.



Crystal 3 YES, this crystal is ideal. The twins are an equal shade of grey when oriented N-S, and the twin sets go into clear extinction at approximately equal angles of clockwise or anticlockwise rotation.



Crystal 4 NO, not appropriate. The crystal has a twin plane down the centre, but the twins are not equal shades of grey when the twin plane is oriented N-S. The crystal is also zoned, which makes it very difficult to determine a unique extinction angle. This optical technique for measuring plagioclase composition does not work well on zoned plagioclases.



Having identified some appropriate crystals in thin section, I measured pairs of extinction angles (clockwise and anticlockwise rotations) needed to put opposite sets of twin planes into extinction. Which pairs of measured extinction angles are suitable for determining plagioclase composition, and which pairs should be rejected?

i)	53° and 57°	Suitable. Average angle = 55°, corresponds to An86
ii)	42° and 41°	Suitable. Average angle = 41.5°, corresponds to An73.
iii)	55° and 38°	Not suitable – angles differ by >5°
iv)	32° and 35°	Suitable. Average angle = 33.5°, corresponds to An61.
v)	21° and 27°	Not suitable – angles differ by >5°

For the plagioclase crystals with pairs of extinction angles that you accepted, calculate the average extinction angle. Then use the diagram on p.27 of the Practical Reference Handout to obtain their compositions.

See above. Allow an uncertainty of about ±2 mol.% An when reading off the graph.

Part 2 (45 minutes): Interpreting basalt crystallization on a ternary phase diagram

Last week we looked at how to describe the process of mantle melting on a ternary phase diagram. This question focuses on the process of crystallization. The logic involved in interpreting the diagram is very similar. Remember that equilibrium crystallization is the exact reverse of equilibrium melting, but fractional crystallization is not the exact reverse of fractional melting. The questions guide you step-by-step through interpreting the crystallization history of a basaltic melt.

The diagram below shows the basalt liquidus surface at 1 atmosphere pressure. The three mineral components are anorthite (An, plagioclase), diopside (Di, clinopyroxene) and forsterite (Fo, olivine) – these plot at the corners of the diagram. The contours show the temperature of the liquidus surface. The solidus is not shown. Use the figure to answer the following questions (you can ignore the spinel field).

1. Melt compositions on this diagram are plotted in terms of their modal abundances (in percent) of Fo, Di and An. What are the modal abundances of Fo, Di and An in bulk composition *x*?

   22% Fo, 53% Di, 25% An. This can be calculated using the lever rule. Allow ±5% for rounding errors.

2. The contours show the temperature of the liquidus. If a melt with bulk composition *x* starts cooling from 1500 °C, at what temperature does it hit the liquidus? (i.e. what is the liquidus temperature for a melt with composition *x*?)

   Point *x* plots between the temperature contours of 1400 and 1500 °C, at about 1430 °C. Any melt with composition *x* that is at a temperature greater than 1430 °C will be completely liquid. As the melt cools from 1500 to 1430 °C its composition does not change, because nothing is crystallizing. Crystallization only starts once the liquidus temperature is reached.

3. When composition *x* hits the liquidus, Fo crystals start to form (this is the definition of the liquidus). The crystallization reaction is L → Fo. As crystallization of Fo proceeds, what will happen to the Mg, Ca and Al concentrations in the melt?

   Fo is Mg-rich olivine, Mg_2SiO_4. As Fo crystallizes, Mg is removed from the melt. Ca and Al do not partition into olivine, so their concentrations in the melt will increase.

4. The changing melt composition as Fo crystallizes is shown by the thick black arrow moving away from *x* in a straight line that extends back towards Fo. The arrow points away from Fo, because Fo crystallization causes the melt composition to be progressively depleted of the chemical ingredients needed to crystallize Fo.

   The black arrow points broadly "downhill" across the temperature contours. (Notice that the arrow is NOT perpendicular to the temperature contours.) At what temperature does the crystallization path change?

   The crystallization path changes when it hits the cotectic (highlighted in red on the diagram). This is quite close to the 1300 °C contour – at around 1320 °C.

   What percentage of the melt has crystallized at the point the crystallization path changes? *(Hint: Use the lever rule. You will need to create a construction line that passes through the composition of the crystallizing mineral, the original bulk composition, and the composition of the liquid at the point the crystallizing path changes. Ask a demonstrator if you're not sure how to do this.)*

   When *x* hits the cotectic, 14% of the original melt has crystallized.

5. The crystallization path changes at the cotectic phase boundary, where Di joins in the crystallising assemblage. Both Fo and Di will now crystallize from the melt; the new crystallization reaction is L → Fo + Di.

   The Fo-Di cotectic is like the bottom of a steep-sided valley, with the cooling liquid constrained by the steep sides to follow the valley down-temperature. Which way is down-temperature on this diagram? Mark this with a down-temperature arrow.

   The down-temperature arrow points towards the ternary eutectic point.

6. When the liquid reaches the eutectic point E at 1270 °C, something new happens. What? Write down the crystallization reaction at the eutectic point.

   At the eutectic point, An starts crystallizing. The eutectic crystallization reaction is L → Fo + Di + An.

7. The eutectic crystallization reaction continues, without the liquid changing composition (because the mixture of crystallizing minerals has the same composition as the liquid), until the last drop of liquid is used up. What minerals are in the final solidified basalt? What are the modal proportions of these minerals in the final solid rock assuming (a) equilibrium crystallization, (b) fractional crystallization?

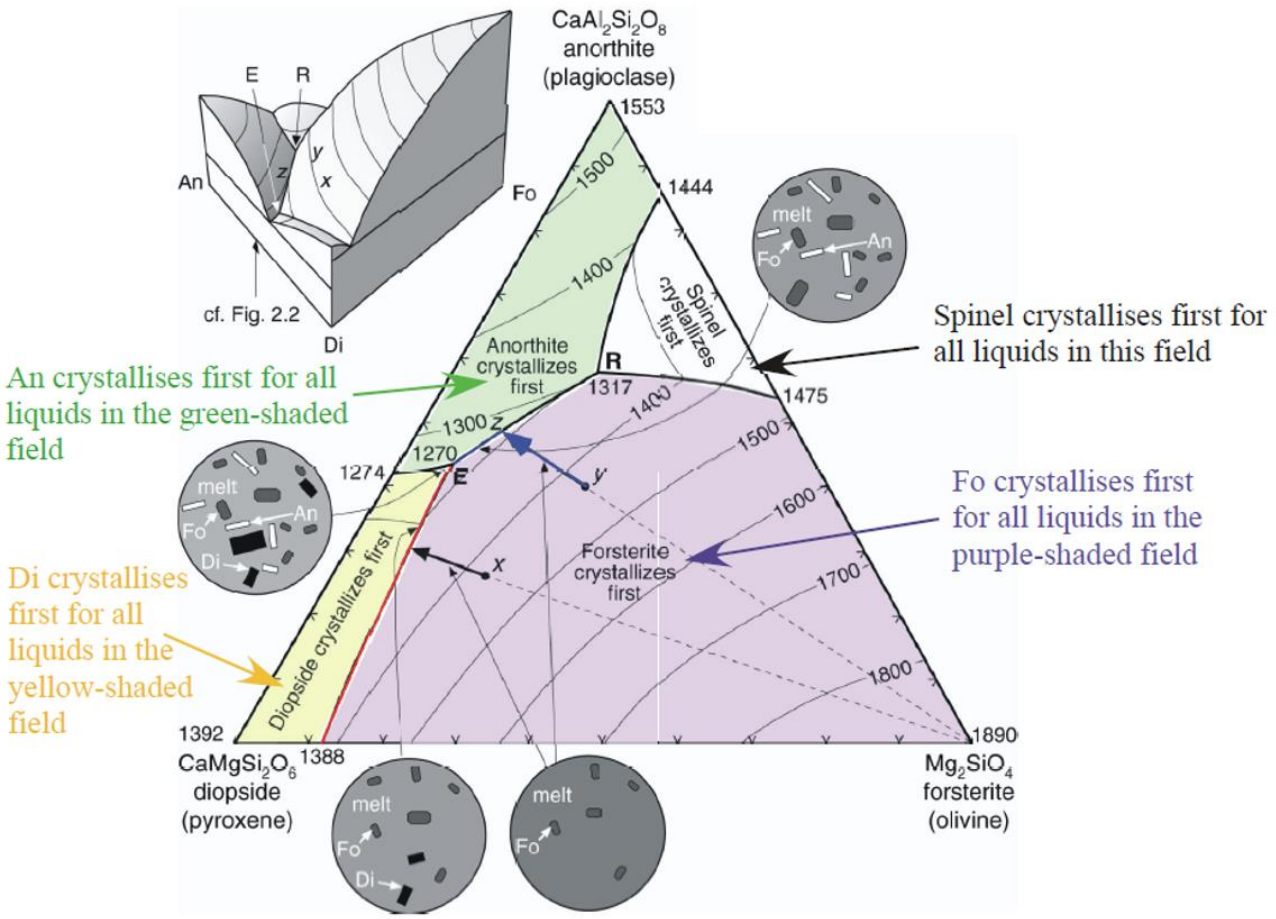
   The final solid contains Fo + Di + An. For equilibrium crystallization, the final solid has the same composition as the starting melt *x* – that is, 22% Fo, 53% Di, 25% An (same as your answer to Q1). For fractional crystallization, the final solid has the same composition as the eutectic, so it will contain 8% Fo, 49% Di, 43% An. Allow ±5% for rounding errors.

8. Use the same steps as described above to describe the crystallization path followed by composition *y*. How is the order of crystallization different from composition x?

   Bulk composition *y* will have the following crystallization sequence:
   - First crystallization reaction in olivine-only field: L → Fo. Melt composition evolves directly away from Fo.
   - Melt composition hits Fo-An cotectic. New crystallization reaction: L → Fo + An. Melt composition evolves along the cotectic boundary in the down-temperature direction.

- Melt composition reaches ternary eutectic. New crystallization reaction: L → Fo + Di + An. Reaction continues until all the liquid is used up.

The order of crystallization for *x* is ol → ol + cpx → ol + cpx + plag.
The order of crystallization for *y* is ol → ol + plag → ol + cpx + plag.



Important take-away points:

- The diagram is a map of the liquidus surface. It can only show liquid compositions.
- Compositions are plotted in terms of the proportions of the phases at the corners of the triangle.
- Temperature is represented by the contours and by the down-temperature arrows.
- During crystallization, liquids can only move down-temperature.
- The last liquid to crystallize will be at the ternary eutectic, E.

Part 3 (60 minutes): A full petrographic rock description **EXAM-STYLE QUESTION**

The aim of this exercise is to produce a full petrographic description of thin section STR or thin section Re114, including sketches. Use the guide to systematic description of igneous thin sections in the Practical Reference Handout p 3-4 to help you. This provides a suggested workflow for the observations you should make, and the level of detail you should aim for.

There will be a question worded in exactly this format on the Practical Test in December. It will be worth **40 marks** (40% of the paper), and in the test you should allocate about 45 minutes to complete this question.

Give a full petrographic description and interpretation of **thin section STR** or **thin section Re114**. Your answer must include at least one labelled diagram illustrating the rock's mineralogy and texture. Marks will be awarded for the following information:

Observations and descriptions of the rock's mineralogy [16]
Observations and descriptions of the rock's texture [10]
Interpretation of the order of crystallization [4]
Name and petrological classification for the rock [4]
Interpretation of the tectonic setting in which this rock may have formed [6]

If you wish to receive **formative feedback** on the petrographic description you produce in today's practical, you may hand in your work at the end of the practical class, and I will return it with some brief written feedback next week.

Individual formative feedback will be provided on any work handed in at the end of the practical class.

General feedback:

The most common suggestions for improvement are:

- Draw the scale bar. Simply writing "x4" (or the magnification of the objective lens) is not enough information.
- Draw neatly. Try to draw the true crystal shapes, rather than schematic blobs.
- Use line thickness to indicate relief.
- Can use colour to draw attention to key features.
- Larger sketches are easier to see and read, and usually contain more detail – so they usually attract higher marks.
- Label all the important minerals and textures. If a feature is important for the interpretation, it needs to be labelled on the diagram.
- If you don't know what a mineral is, describe its optical properties. You can still get high marks for describing a mineral (or a texture), even if you don't get the identification right.
- Use your knowledge of mineral associations to predict groundmass minerals (e.g. basalt should contain cpx and plagioclase in the groundmass).
- Include an estimate of modal proportions - this is often forgotten, but there are marks for it.
- Order of crystallization: fine to say that you cannot tell the order of crystallization – this can be a valid interpretation and will receive marks as long as it is justified.

The "tectonic setting" question is testing your interpretation skills. The most common mistake here is to write "volcanic" or "plutonic" as the tectonic setting, which is not correct (magmas can be extruded or intruded at any tectonic setting). "Tectonic setting" is asking you to decide if you think the magma was formed at a mid-

ocean ridge, or at a hotspot, or at a subduction zone, or at another type of location where magmatic activity occurs. You then need to justify your interpretation. Example: You have a thin section of a basalt that has a glassy groundmass and no vesicles. Your answer to the tectonic setting question might look something like this: "This basalt was probably formed at a mid-ocean ridge. Its glassy groundmass indicates very rapid cooling, but it has no vesicles, which indicates that it erupted under sufficient pressure for gas bubbles not to exsolve. This suggests an underwater eruption, with cold water that is deep enough to prevent a phreatomagmatic explosive eruption. These conditions are found at a mid-ocean ridge."

At this stage of the unit we have not finished discussing magma formation and eruption at all the different tectonic environments, but by the end of the unit you will be able to build these kinds of interpretations from your own knowledge. Even if you don't get the tectonic setting right, you can get high marks for explaining how your interpretation is justified by your observations.

Bonus digital resources: Hand specimens, virtual outcrops and thin sections

Hand specimens

These three samples come from different parts of the oceanic crust. Look for key mineralogical and textural features (or lack of features). The questions for each sample are to guide your observations and encourage you to think about how to interpret what you are seeing.

Deep sea pillow basalt

What evidence is there that this rock cooled rapidly? It is very fine-grained. There is also evidence of a glassy rind around the pillow exterior (the dark shiny patches). Basalt glass only forms if the melt is quenched very quickly.

What do the vesicles represent? They are bubbles of gas (mostly H_2O vapour) that have exsolved from the basalt melt. (They are not trapped bubbles of vapourized seawater.)

Gabbro

What minerals are in this rock? Plagioclase (white) and clinopyroxene (black on the fresh sawn surface, dusty brown-green on the weathered surface). There are probably other minor minerals that can't be seen easily in the hand specimen.

How quickly did this rock cool, and what is the evidence for this? Intermediate to slow cooling. Evidence is the coarse grain size.

Do you think this rock came from the massive gabbros or the foliated gabbros in the oceanic crust, and what is the evidence for this? Probably from the massive gabbros because there is no evidence of a foliation or mineral alignment in the hand specimen.

Serpentinized harzburgite

What evidence is there that fluids have flowed through this rock? Veins, which indicate the fluid flow pathways. The plagioclase vein was originally a melt vein, but fluids could have flowed here too. The many thin lines at ~80° angles are probably fluid flow pathways.

What were the original minerals in this rock? It's a harzburgite, so the original minerals were olivine and orthopyroxene. Green olivine is no longer visible in the hand specimen because it has all been altered to serpentine. The black serpentine gives the rock a micaceous, flaky and shiny appearance. The bright gold patches are orthopyroxene.

Does this rock come from above or below the petrological MOHO? Below. The petrological Moho separates mantle rocks from crustal rocks. This is a mantle rock so it must come from below the petrological Moho.

Virtual outcrops

Pillow basalt, San Francisco Bay
Pillow basalt, Columbia River Basalts, Washington
Can you distinguish individual pillows in the outcrop? The San Francisco outcrop has one distinct pillow shape. The Columbia River pillows are harder to pick out because the edges are less well defined, but the pillows do form blocky masses.
Can you see any evidence for rapid cooling at the pillow edges? If not, why do you think this might be? San Francisco: The black glassy rind is largely gone, but there is a distinctive fine-grained rind around the pillow. Columbia River: The individual pillows are rather eroded and there is no glass preserved at their edges. The resolution of the virtual outcrop is not good enough to see if there are fine-grained rims around the pillows. Why is no glass preserved? Basalt glass is prone to alteration and erosion: it will easily alter to fine-grained clay minerals under the right conditions, and these are easy to weather out. So basalt glass is not often preserved in weathered outcrops.
Both outcrops are very fractured. What do you notice about the orientation of the fractures? The fractures are cooling joints and they form perpendicular to the cooling surface, so they have a radial orientation. Best seen in the Columbia River virtual outcrop.

Thin sections

None of these thin sections is a mid-ocean ridge basalt, but they all show key features of basalts that you should be able to recognise and describe.

Basalt, Staffa
Briefly examine the object (hand specimen) - notice how the weathered exterior of the hand specimen could easily be mistaken for sandstone.
What minerals can you identify in the groundmass of the thin section? Groundmass is very fine. Small plagioclase laths easily recognisable. Clinopyroxene best identified from its light orange interference colours in XPL (it must be cpx because this is an essential mineral in basalts). Unusually, there are small olivines in the groundmass too. There are some patches that appear black/dusty in PPL – these are not glass, but the minerals are too small to identify. There are some green-brown gungy patches, especially along the horizontal vein – this is a secondary mineral called smectite that forms during hydrothermal alteration.
What is the texture of the thin section? No phenocrysts, so the texture is aphyric. The groundmass is holocrystalline (composed entirely of crystals).

S339 01 Basalt
What is the texture of the thin section? Holocrystalline. The texture is not fully interstitial, because all of the constituent minerals have well-developed habits and none of them seems to be filling interstitial spaces. In some places it is possible to see cpx growing around earlier-formed plagioclase – this texture is best described as sub-ophitic.
What interpretations can you make about the order of crystallization? (Rotation 2 is helpful.) Rotation 2 shows olivine fully enclosed by cpx; plagioclase and magnetite fully or partially enclosed by olivine; and plagioclase partially enclosed by cpx. In the wider sample, there are some plagioclase phenocrysts, and the groundmass plagioclases have well-developed tabular habits. The plagioclase phenocrysts grew first. The groundmass minerals probably crystallized at about the same time. Based on the observations, a possible order of crystallization is: plag → plag + ol + mgt → plag + ol + mgt + cpx.
Is there any evidence for secondary alteration? (Rotation 1 is helpful.) Yes – the olivine is altered along crystal edges and along curving cracks through the crystal. But the alteration is not extensive.

Porphyritic olivine basalt, Mauritius

What are the phenocryst minerals in this thin section? Olivine and clinopyroxene.

Are the phenocryst minerals the same as, or different from, the minerals in the groundmass? Different: the groundmass contains plagioclase, cpx and magnetite, but no olivine. Olivine must have finished crystallizing before the groundmass was formed.

What was the likely cooling history of this rock, and what is the evidence for this? Two distinct stages of cooling. Euhedral to subhedral phenocrysts grew free-floating in a magma chamber under slow cooling conditions (low nucleation rate, high growth rate). Fine-grained groundmass formed during rapid cooling (high nucleation rate, low growth rate), probably following a volcanic eruption.

Summary interpretation questions:

Do any of the thin sections preserve evidence of instantaneous melt quenching? No, none of them are vitrophyric. None of them have glass in the groundmass – they are all fully crystalline, even though the groundmass crystals are small.

Which of the three thin sections do you think had the fastest cooling rate, and why? The rock with the finest-grained groundmass cooled fastest. In my opinion the porphyritic basalt from Mauritius is slightly finer, and the groundmass crystals are more acicular/feathery than in the Staffa basalt, so it probably had a faster cooling rate – although there is not much to choose between these two samples. The sub-ophitic basalt has a much coarser groundmass, so it cooled more slowly.